In Search of an Equivariant Characterization of the Tropical Grassmannian $G_{tr}(2,4)$

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April 26, 2018

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- Subspaces and Orbits

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  Real three-dimensional space ($\mathbb{R}^3$) with real scalar multiples is a vector space. The Cartesian Plane ($\mathbb{R}^2$) is a subspace of $\mathbb{R}^3$.

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- **Example**
  Let $V = \mathbb{R}^4$, the Grassmannian $G(2, \mathbb{R}^4)$ is the set of all planes passing through the origin in $\mathbb{R}^4$.

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- (Equivariant) Quotients of a group acting on the set

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- The kth **exterior power** $\bigwedge^k V$ of an n-dimensional vector space V is a space whose elements are the kth exterior products of elements of V. i.e. for $v_1, \ldots, v_k \in V$, $v_1 \wedge \cdots \wedge v_k \in \bigwedge^k V$.

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The **Plücker Embedding** is a map $\Psi : G(k, V) \to \mathbb{P}(\bigwedge^k V)$ defined by $\text{span}(v_1, \ldots, v_k) \mapsto [K(v_1 \wedge \cdots \wedge v_k)]$

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- These end up being the coefficients of a family of polynomials whose common zero locus is a line in $\mathbb{P}(\bigwedge^k V)$.

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- The **stabilizer** of a subset $X \subset S$, $\mathrm{Stab}(X)$, is the subgroup

$$\{g \in G \mid \forall x \in X,\ g(x) \in X\}$$

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- The set of invertible $n \times n$ matrices over a field F is a group $GL(n, F)$.
- This group acts on a vector space V over a field F through good ol' matrix-vector multiplication.
- Given a k-dimensional subspace $K \subset V$, it is the span of k independent vectors $v_1, \ldots, v_k$. Thus, $\mathrm{Stab}(K)$ is just the group of invertible matrices that preserve the basis vectors $v_1, \ldots, v_k$ up to permutations and scalar multiples.

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  - Then $b^{-1}a(K) = b^{-1}(a(K)) = b^{-1}(Q) = K$, which means $b^{-1}a \in H$, so $\exists h \in H$ such that $b^{-1}a = h$.

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  - Then $b^{-1}a(K) = b^{-1}(a(K)) = b^{-1}(Q) = K$, which means $b^{-1}a \in H$, so $\exists h \in H$ such that $b^{-1}a = h$.
  - Then we have $a = bh$ for some $h \in H$.
  - Therefore, $aH = (bh)H = bH$.

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Which do you prefer?

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- These are equivalent, but I personally prefer the first one.

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The Tropical Semiring

$(\mathbb{R} \cup \{-\infty\}, \oplus, \odot)$ is the Tropical Semiring!

Speyer and Sturmfels: Tropical Grassmannian

- A **tropical hyperplane** $\mathcal{H}(\ell)$ where ℓ is a tropical linear form in n-unknowns:

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- We consider the $\binom{n}{d}$-dimensional space $\mathbb{R}^{\binom{n}{d}}$ whose coordinates $X_{i_1 \ldots i_d}$ we index by d-element subsets $\{i_1, \ldots i_d\} \subset \{1, 2, \ldots, n\}$.

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- Let S be any (d-2)-element subset of $\{1, 2, \ldots, n\}$ and let i, j, k, l be any four distinct indices in $\{1, 2, \ldots, n\} \backslash S$.
- We can then describe the **three-term Grassmann-Plücker relation** via the general polynomial

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- They then define the **three-term Tropical Grassmannian** to be the intersection

$$G_{tr} = \bigcap_{S,i,j,k,l} \mathcal{H}(p_{S,i,j,k,l}) \subset \mathbb{R}^{\binom{n}{d}}$$

Research Problem

In classical geometry, the equational parametrization is equivalent to a nice group theoretic parametrization. Is this the case for the Tropical Grassmannian?

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$$(u_1, \ldots, u_n) \oplus (v_1, \ldots, v_n) = (v_1 \oplus u_1, \ldots, v_n \oplus u_n)$$

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- Define scalar multiplication to be

$$\lambda \odot (v_1, \ldots, v_n) = (\lambda \odot v_1, \ldots, \lambda \odot v_n)$$

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- Let $\mathbb{R}^n_{\max}$ be the set $\mathbb{R} \cup \{-\infty\}$
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$$(u_1, \ldots, u_n) \oplus (v_1, \ldots, v_n) = (v_1 \oplus u_1, \ldots, v_n \oplus u_n)$$

- Define scalar multiplication to be

$$\lambda \odot (v_1, \ldots, v_n) = (\lambda \odot v_1, \ldots, \lambda \odot v_n)$$

- This doesn't satisfy all the criteria for a proper vector space (a module over a field), but behaves enough like a one for us to treat it as such.

Tropical General Linear Group

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- Example

  Some examples from $\mathbb{M}^3$:

$$\begin{pmatrix} -5 & -\infty & -\infty \\ -\infty & 7 & -\infty \\ -\infty & -\infty & 3 \end{pmatrix}, \begin{pmatrix} -\infty & -10 & -\infty \\ 14 & -\infty & -\infty \\ -\infty & -\infty & 4 \end{pmatrix}, \begin{pmatrix} -\infty & -\infty & 0 \\ -\infty & 0 & -\infty \\ 0 & -\infty & -\infty \end{pmatrix}$$

Tropical General Linear Group

Definition

Tropical matrix multiplication of $n \times n$ matrices is defined by

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \times \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nn} \end{pmatrix} =$$

$$\begin{pmatrix} a_{11} \odot b_{11} \oplus \cdots \oplus a_{1n} \odot b_{n1} & \cdots & a_{11} \odot b_{1n} \oplus \cdots \oplus a_{1n} \odot b_{nn} \\ \vdots & \ddots & \vdots \\ a_{n1} \odot b_{11} \oplus \cdots \oplus a_{nn} \odot b_{n1} & \cdots & a_{n1} \odot b_{1n} \oplus \cdots \oplus a_{nn} \odot b_{nn} \end{pmatrix}$$

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  - Then for fixed i, $\exists!$ j such that $(m_{i,j}) \neq -\infty$. Similarly for fixed j.
  - Therefore, the product of monomial matrices is a monomial matrix.

Tropical General Linear Group

- Associativity
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  - Thus, $AA^{-1} = \mathbb{I}$.

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Tropical General Linear Group

Thus, $\mathbb{M}^n$ is a group under tropical matrix multiplication.

- Cool, but what should our subspaces be?
- Let's try solutions to tropical polynomial equations.

Subspaces of $\mathbb{R}^4_{\max}$ and Orbits

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- Definition

  A 2-dimensional **tropical subspace** of $\mathbb{R}^3_{\max}$ is the set of solutions to the general tropical polynomial:

$$A \odot x \oplus B \odot y \oplus C \odot z = 0; \qquad A, B, C \in \mathbb{R} \cup \{-\infty\}$$

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- When one of $A, B, C = -\infty$, the result is two-faced plane with a free variable corresponding to the infinite coefficient.
- When two of $A, B, C = -\infty$, the result is a single faced plane with two free variables corresponding to the infinite coefficient.

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- Moreover, $\forall x \in \mathbb{K},\ \lambda x \notin \mathbb{K}$
- Thus, this cannot be a characterization of 2-dimensional subspaces of $\mathbb{R}^3_{\max}$.

Subspaces of $\mathbb{R}^4_{\max}$ and Orbits

So let's try something else...

Subspaces and Orbits

So let's try something else:

Definition

Define the **span of tropical vectors** $(u_1, \ldots, u_n)$ and $(v_1, \ldots, v_n)$ to be

$$(\lambda_u \odot u_1 \oplus \lambda_v \odot v_1, \ldots \lambda_u \odot u_n \oplus \lambda_v \odot v_n)$$

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#	Characterization	Example
i	$u_1 = u_2 = -\infty$	$(-\infty, -\infty, 0), (2, 3, 0)$
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- Sort of strange, but it works.

Subspaces of $\mathbb{R}^4_{\max}$ and Orbits

We can quickly try to visualize what these might look like:

Example

- Let $\vec{a} = (2, 5, 0)$ and $\vec{b} = (8, -5, 0)$.

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- By restricting our choices of λ_a and λ_b, we can view the region of $\text{span}(\vec{a}, \vec{b})$ at $z = 0$.

$$\text{span}(\vec{a}, \vec{b}) = (\lambda_a \odot 2 \oplus \lambda_b \odot 8, \lambda_a \odot 8 \oplus \lambda_b \odot -5, \lambda_a \oplus \lambda_b)$$

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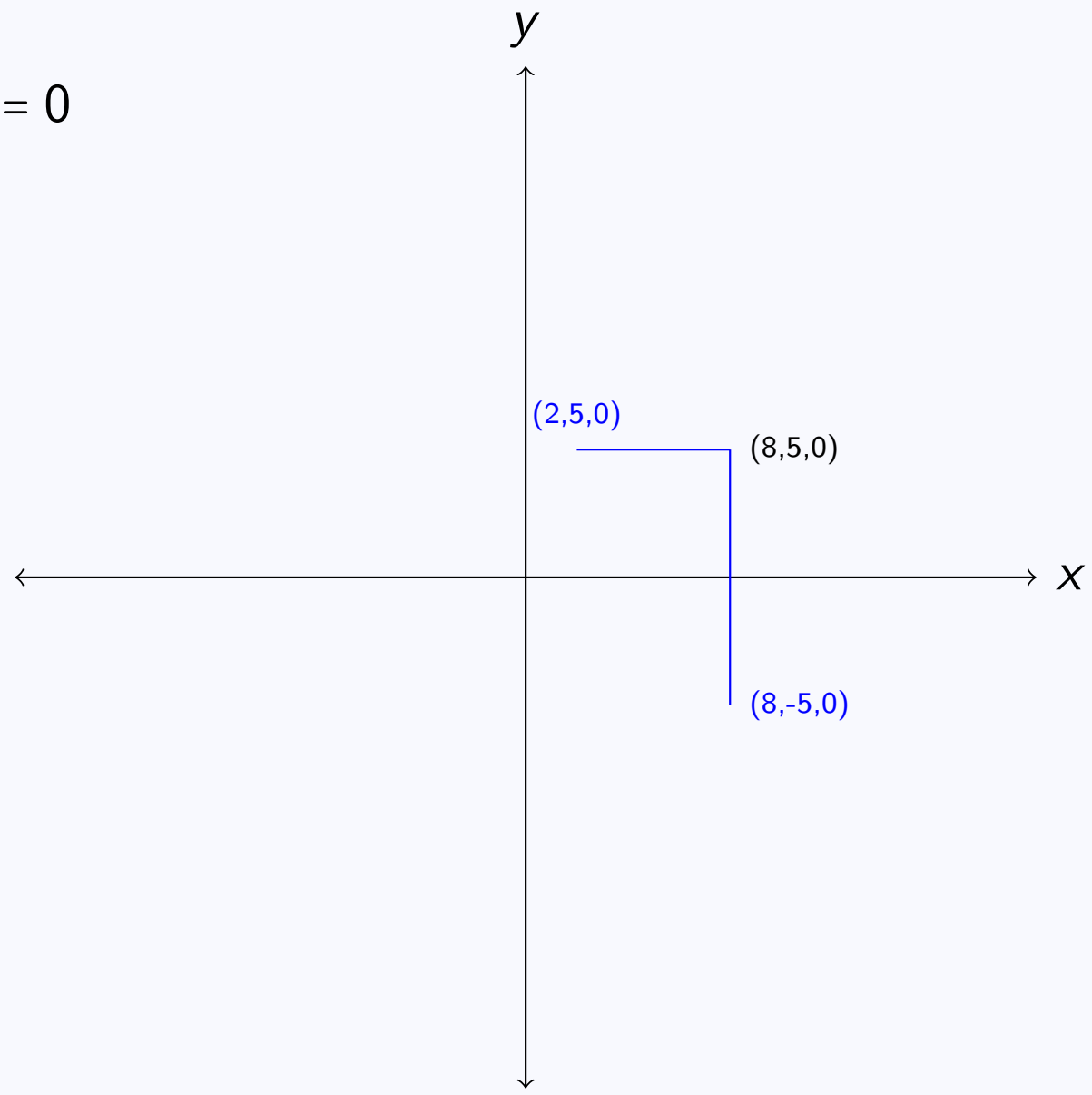
- To remain within $z = 0$, we need $\lambda_a \oplus \lambda_b = 0$.
- If $\lambda_a = 0$, $\lambda_b \leq 0$, then $y = 5$ while $2 \leq x \leq 8$.
- If $\lambda_a \leq 0$, $\lambda_b = 0$, then $x = 8$ and $-5 \leq y \leq 5$.

$z = 0$



Subspaces of $\mathbb{R}^4_{\max}$ and Orbits

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- Through the same calculations performed earlier, we create the graph...

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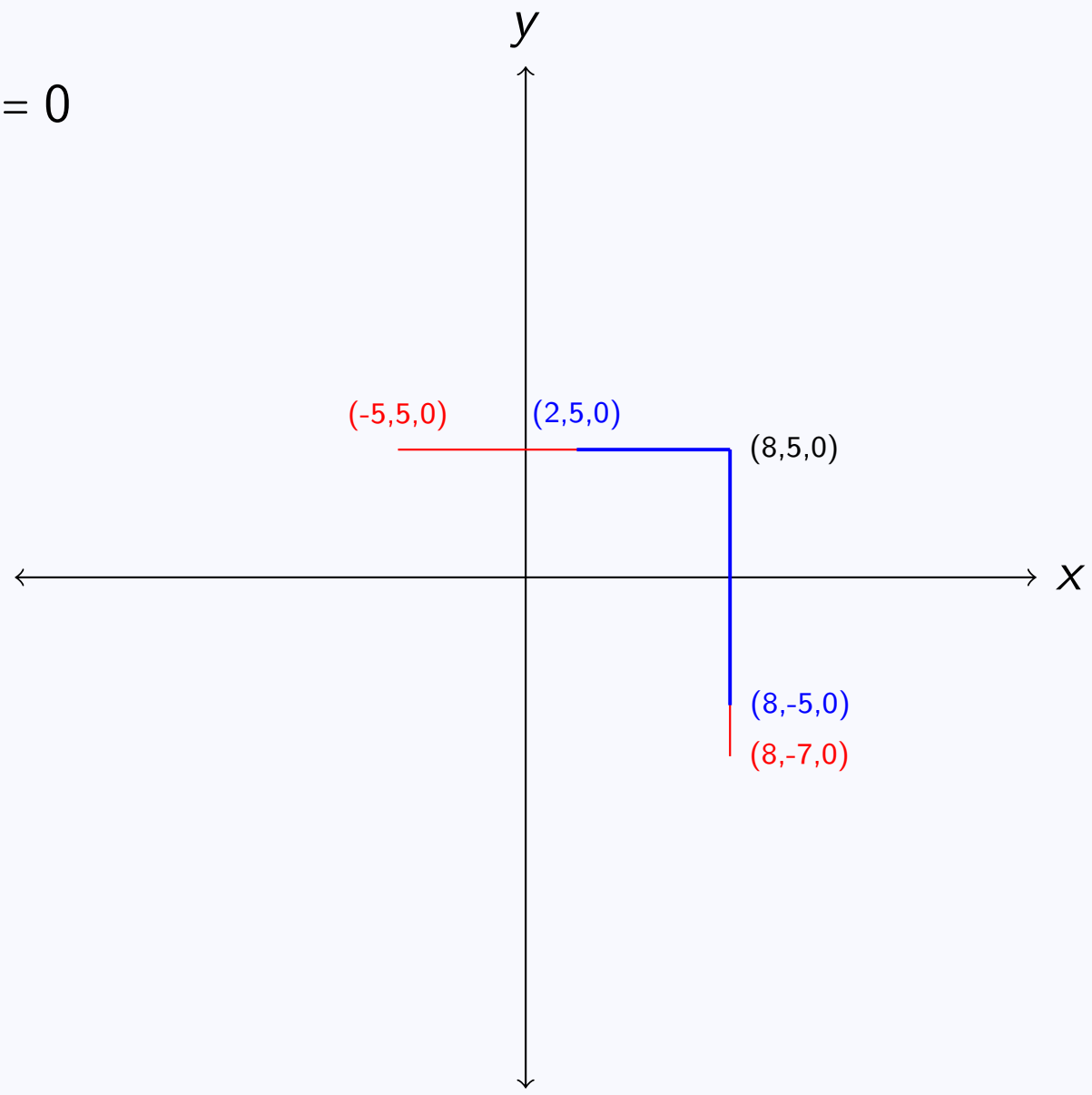

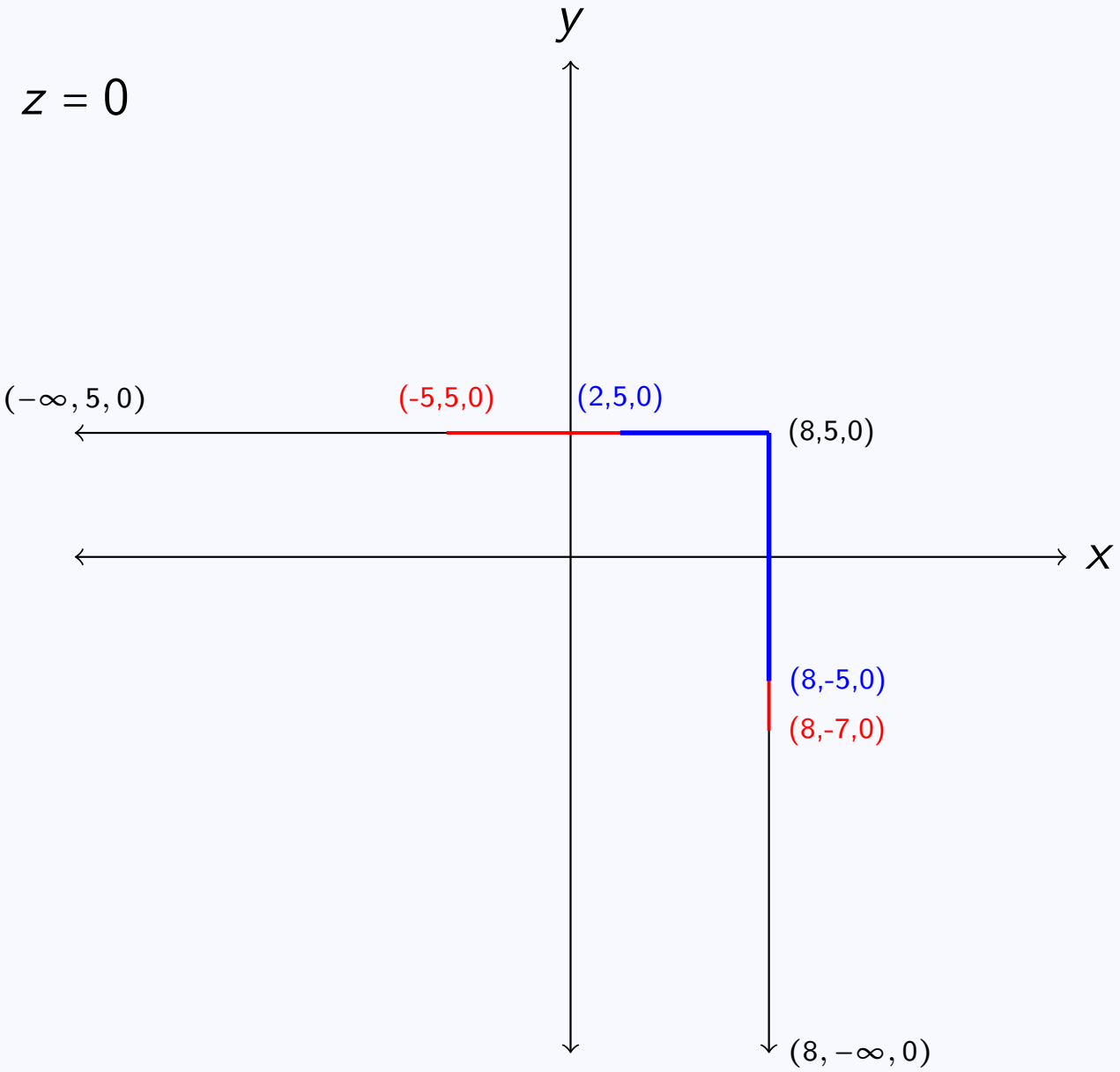
$z = 0$
y
x
$(-\infty, 5, 0)$
(-5,5,0)
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Subspaces of $\mathbb{R}^4_{\max}$ and Orbits

- Given an arbitrary two-dimensional subspace S, can we parametrize $G_{tr}(2,4)$ by taking the quotient of $\mathbb{M}^4$ by $\mathrm{Stab}(S)$?

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    - The proof of this basically amounts to showing that our monomial matrices cannot change pair-wise orderings or produce $-\infty$ coordinates in tropical vectors.

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- We can, however, determine which types are orbits under group action by monomial matrices.
- It turns out that ii, iii, and iv are orbits.
- i and v are not orbits

Closing Remarks and Further Work

Future work on this might try to answer the following:

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- In n-dimensional tropical vector space, given some integer $0 < k < n - 1$, how many k-dimensional subspace types are there? How many of these are orbits?

Closing Remarks and Further Work

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- Is an equivariant characterization of Sturmfels' and Speyer's equational characterization possible? If not, is there a partial equivariant characterization that parametrizes subspaces which are orbits?

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Thank you for listening.
Special thanks to my research advisor
Dr. Murphy.